

clustering

1. Apply KMeans algorithm to cluster the following four objects (with (x, y) representing locations) into two clusters. Initial cluster centers are: Medicine A (1, 1) and Medicine B (2, 1). Use Euclidean distance. Explain each of the clustering steps and the clustering result after each iteration.

Object	Attribute 1 (x)	Attribute 2 (y)
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

2. Choose an appropriate data set **other than Iris data set** and apply K-means to cluster the data using **R and Weka**. You should **use the same data set** and **set the same number of clusters** for both tools. Describe the clustering results generated from these two tools, compare them, and explain which result is better than the other.

You also need to submit the screenshots showing the steps of performing clustering using R and Weka and the visualization of the clustering results.

References:

- a) For R, you can find documentation and examples here:

<http://www.rdatamining.com/examples/kmeans-clustering>

- b) A document describing how to cluster using Weka is posted on myCourse under **Homework** Section.